

SmartCode Production Owner Handbook

Practical operating guide for running, debugging, and improving SmartCode in production. Current live website: <https://smartcode-three.vercel.app>

Hosting stack

Frontend on Vercel, backend on Render, persistent data in MongoDB Atlas.

Core priority

Keep signup, login, problem listing, run, submit, and leaderboard stable first.

Daily operating routine

- Open the live site and test homepage, login, and one problem route every morning.
- After each deploy, manually test signup, login, problem run, and problem submit.
- Review Render logs once daily for repeated crashes, startup errors, or route failures.
- Review Vercel deployment status after every push to main.

Weekly operating routine

- Check MongoDB Atlas metrics for connections, storage growth, and query health.
- Review bug reports and classify them as frontend, backend, or data issues.
- Export or snapshot important data before any structural or schema change.
- Verify production environment variables still match the live domains and current database credentials.

Launch and post-deploy checklist

Step	Action	Success signal
1	Push code and confirm GitHub has the intended commit.	Correct branch and commit message visible.
2	Let Render redeploy the backend.	No crash in Render logs.
3	Let Vercel redeploy the frontend.	Deployment completes successfully.
4	Smoke-test the live site.	Auth and coding flow work end to end.
5	Check browser console and network tab.	No CORS or failed API calls.

Where to debug first

Symptom	Check first	Likely cause
Frontend loads but actions fail	Browser network tab	Bad API base URL, CORS issue, backend route error
Login or signup fails	Render logs and Atlas connection	Database auth failure or server-side validation issue
Slow first request	Render service behavior	Cold start on lower backend tier
Problem run or submit fails	Backend logs around judge routes	Judge dependency, route error, or bad payload

Incident runbooks

- If login stops working: inspect the request in browser devtools, then open Render logs and verify MongoDB Atlas connectivity.
- If CORS errors appear: confirm Render CORS_ORIGIN exactly matches the Vercel domain including protocol.
- If backend deploy fails: read the first stack trace in Render logs, then confirm root directory, dependencies, and environment variables.
- If the whole site feels slow: compare browser timing, Render logs, and Atlas metrics to isolate frontend, backend, or database latency.

Data and change safety

Before risky changes

Take a backup or export, test on a safe dataset, then deploy small changes.

Known risk to fix next

Passwords are currently plain text. Move to bcrypt before broader public growth.

Core tools

- Vercel dashboard for frontend builds and environment variables.
- Render dashboard for backend logs, deploy history, and runtime settings.
- MongoDB Atlas for data metrics, database users, backups, and credential rotation.
- GitHub for change history, rollback, and release traceability.